

Project work, part 4 - Machine Learning -2

Tasks

Jupyter Notebook

I saw that production data has 3,647 fewer rows than consumption data.

I padded the missing data with NaN values (specifically for NO5 and wind, covering 2021-01-01 to 2021-06-01).

```
In [2]: from utils.data_loaders import load_mongoDB

MONGO_DATABASE = "elhub_data"
MONGO_COLLECTION_PRODUCTION_FULL = "production_data_2021_2024"
MONGO_COLLECTION_CONSUMPTION_FULL = "consumption_data_2021_2024"

df_consumption = load_mongoDB(MONGO_COLLECTION_CONSUMPTION_FULL, MONGO_DATABASE)
df_consumption.head()
df_consumption.tail()
```

```
2025-11-19 07:22:06.042 WARNING streamlit.runtime.caching.cache_data_api: No runtime
found, using MemoryCacheStorageManager
2025-11-19 07:22:06.042 WARNING streamlit.runtime.caching.cache_data_api: No runtime
found, using MemoryCacheStorageManager
2025-11-19 07:22:06.047 WARNING streamlit.runtime.scriptrunner_utils.script_run_cont
ext: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when
running in bare mode.
2025-11-19 07:22:06.219
Warning: to view this Streamlit app on a browser, run it with the following
command:

streamlit run c:\Users\oriek\miniconda3\envs\D2D_env\lib\site-packages\ipykernel
_launcher.py [ARGUMENTS]
2025-11-19 07:22:06.220 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:06.222 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:06.222 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:06.789 Thread 'Thread-3': missing ScriptRunContext! This warning ca
n be ignored when running in bare mode.
2025-11-19 07:22:06.804 Thread 'Thread-3': missing ScriptRunContext! This warning ca
n be ignored when running in bare mode.
2025-11-19 07:22:06.804 Thread 'Thread-3': missing ScriptRunContext! This warning ca
n be ignored when running in bare mode.
2025-11-19 07:22:39.623 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:39.642 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:39.644 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
```

```
Out[2]:
```

	pricearea	datatype	groupname	starttime	quantitykwh
876595	NO5	Consumption	cabin	2024-04-29T09:00:00+02:00	31118.297
876596	NO5	Consumption	household	2024-10-26T22:00:00+02:00	390373.800
876597	NO5	Consumption	secondary	2024-04-20T14:00:00+02:00	1031581.400
876598	NO5	Consumption	household	2024-03-14T08:00:00+01:00	485459.100
876599	NO5	Consumption	tertiary	2023-09-24T10:00:00+02:00	234220.220

```
In [3]: df_production = load_mongoDB(MONGO_COLLECTION_PRODUCTION_FULL, MONGO_DATABASE)
df_production.head()
df_production.tail()
```

```
2025-11-19 07:22:42.915 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:42.916 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:42.916 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:42.917 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:22:43.426 Thread 'Thread-4': missing ScriptRunContext! This warning ca
n be ignored when running in bare mode.
2025-11-19 07:22:43.426 Thread 'Thread-4': missing ScriptRunContext! This warning ca
n be ignored when running in bare mode.
2025-11-19 07:22:43.429 Thread 'Thread-4': missing ScriptRunContext! This warning ca
n be ignored when running in bare mode.
2025-11-19 07:23:28.305 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:23:28.306 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
2025-11-19 07:23:28.306 Thread 'MainThread': missing ScriptRunContext! This warning
can be ignored when running in bare mode.
```

```
Out[3]:
```

	pricearea	datatype	groupname	starttime	quantitykwh
872948	NO5	Production	solar	2023-10-14T08:00:00+02:00	10.396
872949	NO5	Production	solar	2024-11-25T12:00:00+01:00	189.010
872950	NO5	Production	solar	2024-04-29T02:00:00+02:00	88.424
872951	NO5	Production	thermal	2024-06-15T09:00:00+02:00	15172.930
872952	NO5	Production	thermal	2024-05-28T16:00:00+02:00	18604.510

```
In [4]: cons_per_time = df_consumption.groupby("starttime").size()
prod_per_time = df_production.groupby("starttime").size()

cons_per_time.describe(), prod_per_time.describe()
```

```
Out[4]: (count      35064.0
mean         25.0
std           0.0
min          25.0
25%          25.0
50%          25.0
75%          25.0
max          25.0
dtype: float64,
count      35064.000000
mean        24.895990
std          0.305278
min          24.000000
25%          25.000000
50%          25.000000
75%          25.000000
max          25.000000
dtype: float64)
```

```
In [21]: actual_per_group = (
    df_production[df_production["pricearea"] == "N05"]
    .groupby("groupname")
    .size()
)

missing_per_group = expected_per_group - actual_per_group
missing_per_group

summary = pd.DataFrame({
    "expected_rows": expected_per_group,
    "actual_rows": actual_per_group,
    "missing_rows": missing_per_group,
    "missing_percent": (missing_per_group / expected_per_group * 100).round(3)
})

print(summary.sort_values("missing_rows", ascending=False))
```

	expected_rows	actual_rows	missing_rows	missing_percent
groupname				
wind	35064	31417	3647	10.401
hydro	35064	35064	0	0.000
other	35064	35064	0	0.000
solar	35064	35064	0	0.000
thermal	35064	35064	0	0.000

```
In [ ]: import pandas as pd

full_index = pd.date_range(
    start="2021-01-01 00:00:00",
    end="2024-12-31 23:00:00",
    freq="h",
    tz="Europe/Oslo"
)

template = pd.DataFrame({"starttime": full_index})

df_no5_wind = df_production[
    (df_production["pricearea"] == "N05") &
    (df_production["groupname"] == "wind")
].copy()

df_no5_wind["starttime"] = pd.to_datetime(df_no5_wind["starttime"], utc=True)
df_no5_wind["starttime"] = df_no5_wind["starttime"].dt.tz_convert("Europe/Oslo")

df_no5_wind_full = template.merge(
    df_no5_wind,
    on="starttime",
    how="left"
)

df_no5_wind_full["pricearea"] = "N05"
df_no5_wind_full["datatype"] = "production"
df_no5_wind_full["groupname"] = "wind"

df_production_filled = pd.concat([
```

```

df_production[
    ~(
        (df_production["pricearea"] == "N05") &
        (df_production["groupname"] == "wind")
    )
], # all other data unchanged
df_no5_wind_full
])

# df_production_filled = df_production_filled.sort_values("starttime")

print(df_production_filled)
print(len(df_production_filled))

```

	pricearea	datatype	groupname	starttime	quantitykwh
0	N01	Production	hydro	2022-01-19T02:00:00+01:00	1500431.000
1	N01	Production	hydro	2021-09-11T21:00:00+02:00	1868775.200
2	N01	Production	hydro	2021-10-26T02:00:00+02:00	2118256.200
3	N01	Production	other	2021-06-09T08:00:00+02:00	6.000
4	N01	Production	wind	2021-05-10T02:00:00+02:00	48078.945
...	...	...	...	...	...
35059	N05	production	wind	2024-12-31 19:00:00+01:00	0.000
35060	N05	production	wind	2024-12-31 20:00:00+01:00	0.000
35061	N05	production	wind	2024-12-31 21:00:00+01:00	0.000
35062	N05	production	wind	2024-12-31 22:00:00+01:00	0.000
35063	N05	production	wind	2024-12-31 23:00:00+01:00	0.000

[876600 rows x 5 columns]
876600

Streamlit app

Refactoring

- Plotting
 - Exchanged static plots (matplotlib) with dynamic plots (Plotly)
- Structure and navigation
 - On the start page, the user must first select an area and the type of energy data to analyze (e.g., Production or Consumption). To ensure consistency across all dependent visualizations and analyses, the user is not allowed to reselect these primary parameters once chosen. A change in the selected area or data type requires the user to explicitly click the "Reset" button located on the top of the page.
 - Energy Visualization & Decomposition, Weather Exploration & Outlier have common data selection. All other application modules have independent date selection.

Observation Meteorology and energy production

- After implementation, play with the controls and check if you can spot any changes incorrelations in normal conditions and in/after extreme weather events. I could not find

such incorrelations.

Bonus content

I have tried to implement the following contents:

- Waiting time
 - Use progress bars, spinners, or similar to indicate work in progress.
 - Cache everything that is possible to cache.
- Error handling
 - Try to incorporate checks in the app that handle missing data connections (API and database) and NaN/missing values in the data.
 - Catch the errors and give useful feedback instead of crashing or giving cryptic error messages.
- Forecasting
 - Add weather properties to the list of exogenous variables and download when needed.

Log Describing the Compulsory Work

Coding in Jupyter Notebook

- I used the code from previous projects, and it went relatively smoothly.
- As I mentioned in the beginning of the report, I created/modified some functions to handle both production and consumption data, and a new table/data structure to accommodate the new functions.

Coding in Streamlit app

1. Exchange matplotlib figures with dynamic plots by plotly

With the help of AI tools, I exchanged all the matplotlib figures for plotly plots in the Jupyter Notebook environment before merging the code into the Streamlit app.

2. All new functions

I modified all new functions, including machine learning (lagged_correlation_plot), SARIMAX, map_folium_choropleth, and others. All the new functions were tested in the Jupyter Notebook environment before merging them into the Streamlit app.

3. Structure and navigation before and under coding

I thought about the necessary input and output, what the user should or should not select, the ranges the user selection covers, and the user experience (e.g., speed, intuitiveness). Since I did not have an overview and understanding of what Streamlit could provide me/us, it was challenging to determine what I could serve the users and how to tune the Streamlit app.

4. Coding, testing, debugging of the entire project

I have one start/home page, seven subpages, and twelve utility files (functions, etc.). It is easy to lose track of the overview. Small changes in one file can have big consequences in other files.

5. Access to remote MongoDB Atlas has become slow

As I mentioned, this happened suddenly. After testing the connections with some help and advice from AI tools, I decided to install MongoDB locally on my PC to continue developing the Streamlit app.

GitHub

I pushed files to GitHub, but the Streamlit app does not work online or, if it does, it runs extremely slowly.

Brief Description of AI Usage

I used AI tools to generate initial versions of the code, to refine and customize them, and to debug errors.

Coding and Debugging

I described my requirements, and AI tools suggested initial drafts. While some of the generated code worked well, others required some modification.

Structure and navigation, testing, debugging of the entire project

This was the most challenging area in this project. When an error occurred, AI tools would tell me where and how I should modify the code. As soon as I made the change, other parts of the project would stop working. This happened repeatedly.

I used AI tools to check the overall consistency of the entire project. The tools cleaned up my code by: checking for lines defined twice, ensuring the use of `st.session_state` was consistent throughout the project, and suggesting the creation of new functions.

When I did not inform the AI tools of my own modifications, they tended to stay within their own understanding and context, without incorporating my ideas and changes. AI tools clearly require context to answer me effectively. The amount of detail regarding my ideas and activities that should be shared with the AI tools remains a key question.